

The Lehmer–Clements enumerator

rank-LCM pruning for prime-complete consecutive pairs,
and the collapse of the gapless search space

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Abstract

The Lehmer–Clements algorithm enumerates prime-complete consecutive pairs $m(m+1)$ of a given order ω by solving Pell equations over the squarefree divisors of the ω -th primorial and pruning, by an exact rank-of-apparition condition, every discriminant that cannot carry the full set of primes $2, 3, \dots, p_\omega$. It refines the classical Størmer–Lehmer enumeration of smooth pairs to the gapless (prime-complete) case. Beyond the role of a faster certified verifier, the algorithm exposes a striking empirical regularity: the number of discriminants that survive the rank pruning—the discriminants that actually require a Pell index check—decreases monotonically as ω grows, from 71 at $\omega = 8$ to 2 at $\omega = 17$, even as the total number of discriminants grows as 2^ω . The survival *rate* falls faster than the discriminant count rises. We document the algorithm, its pruning gates, and this collapse, and we isolate the resulting open problem: a bound on the number of small-rank-LCM discriminants, which would close the prime-complete termination question by a combinatorial route rather than an analytic one.

1 Setup

For $N = m(m+1)$ with $\gcd(m, m+1) = 1$, the prime factors of m and $m+1$ are disjoint. Write $p_1 < p_2 < \dots$ for the primes, $P_\omega = \prod_{i \leq \omega} p_i$ for the ω -th primorial. A product N is *prime-complete* of order ω if $\text{rad}(N) = P_\omega$, i.e. the primes dividing N are exactly $\{p_1, \dots, p_\omega\}$ with no gaps.

By Størmer’s theorem each p_ω -smooth consecutive pair corresponds to a solution of a Pell equation. Writing $4m(m+1) = x^2 - 1$ with $x = 2m+1$, a pair of order dividing ω arises from

$$x^2 - D y^2 = 1, \quad D \mid P_\omega \text{ squarefree},$$

where D is the squarefree kernel contributed to the pair. The solutions are the powers $(x_1 + y_1 \sqrt{D})^j$ of the fundamental solution; we write (x_j, y_j) for the j -th, and $m_j = (x_j - 1)/2$. Lehmer’s primitive-divisor analysis bounds the relevant indices: every p_ω -smooth pair occurs at an index

$$j \leq L(\omega) = \max(30, p_\omega + 1).$$

2 The two pruning gates

The enumerator visits each squarefree $D \mid P_\omega$, computes the fundamental (x_1, y_1) , and applies two exact gates before any large-index work.

Gate 1: y_1 -smoothness (Lucas divisibility). For every j , $y_1 \mid y_j$ in the Pell–Lucas sequence. Hence if y_1 is not p_ω -smooth, no y_j is smooth, and D can produce no smooth pair. Such D are discarded. This gate removes the large majority of discriminants (e.g. 63,491 of 65,535 at $\omega = 16$).

Gate 2: rank-LCM (the Lehmer–Clements condition). Fix D passing Gate 1. For the pair at index j to be prime-complete, every prime $p \leq p_\omega$ not already supplied by D or by y_1 must divide $m_j(m_j + 1)$. For an unramified odd prime p ($p \nmid D$), $p \mid y_j$ iff j is a multiple of the *rank of apparition*

$$\rho_p(D) = \min\{j \geq 1 : p \mid y_j\},$$

computed by iterating the recurrence modulo p up to $L(\omega)$. For the pair to carry *all* missing primes, j must be a common multiple of every required $\rho_p(D)$; hence j must be a multiple of

$$R(D) = \text{lcm}\{\rho_p(D) : p \text{ missing}\}.$$

Two outcomes prune D outright:

1. some required prime p has no rank within the certified window ($\rho_p(D)$ undefined for $j \leq L$): then p can never appear, and D is dead (RANK_MISSING);
2. $R(D) > L(\omega)$: then no admissible index is a multiple of $R(D)$, so no prime-complete pair occurs (LCM_EXCEEDS_L).

Only when $R(D) \leq L$ does D *survive*, and then only the multiples $R(D), 2R(D), \dots \leq L$ need a full smoothness check. The missing primes are processed largest-first, since large primes tend to force large ranks and trip the $R > L$ cutoff early.

Remark 1 (Why this is exactly the gapless refinement). The classical Lehmer enumeration checks every smooth pair. Gate 2 adds the prime-complete requirement directly: a prime of *zero* exponent in $m_j(m_j + 1)$ —a gap—is forbidden, and the rank condition is precisely the arithmetic statement that no required prime is missing at the admissible indices. Pruning a discriminant by $R > L$ is pruning a branch of the search tree on which some prime would necessarily have zero exponent.

3 Soundness and verification

The verdict “no prime-complete pair at order ω ” is unconditional given the index ceiling $L(\omega)$: the rank-LCM condition is a necessary condition on any prime-complete index, and all indices $\leq L$ are exhausted. The implementation gates every certificate run through a regression self-test that recovers all 28 catalogued prime-complete pairs for $\omega \leq 8$ (A141399); certify mode refuses search caps. Across $\omega = 8, \dots, 16$ the accounting balances exactly (every discriminant is classified into prefilter, Gate 1, Gate 2, or a checked index), and the run reproduces the single known pair 633555 at $\omega = 8$ and finds none above.

4 The collapse of the survivor set

The quantity of interest is $S(\omega)$, the number of discriminants surviving both gates (status RANK_SURVIVES), and $C(\omega)$, the total Pell indices then checked. The measured values:

ω	total $D = 2^\omega - 1$	Gate 1 removed	survivors $S(\omega)$	indices $C(\omega)$
8	255	112	71	236
9	511	287	66	194
10	1023	695	53	114
11	2047	1573	47	91
12	4095	3442	30	51
13	8191	7310	26	41
14	16383	15198	9	11
15	32767	31189	4	10
16	65535	63491	3	3
17	131071	128443	2	5

$S(\omega)$ is strictly decreasing—71, 66, 53, 47, 30, 26, 9, 4, 3, 2—with no reappearance, while the total discriminant count grows as 2^ω . A least-squares fit over $\omega = 8, \dots, 16$ gives

$$S(\omega) \approx \exp(-0.42\omega + 8.1),$$

a halving roughly every 1.6 levels, with the survival *rate* $S(\omega)/2^\omega$ falling from 0.28 at $\omega = 8$ to 4.6×10^{-5} at $\omega = 16$. The $\omega = 17$ value is a genuine out-of-sample test: the fit, formed without it, predicts $S(17) \approx 2.4$, and the measured value is 2. Only $\omega = 8$ produces a smooth pair at all (the terminal 633555); every later level checked has $C(\omega)$ indices and zero smooth pairs.

The race in logarithms. Per unit increase in ω , the discriminant count contributes $+\log 2 \approx +0.69$ to $\log S$, while the empirical survival rate contributes about -1.11 , for a net -0.42 . The rank-LCM survival rate decays faster than the discriminant count grows; the margin 0.42 per level is what drives $S(\omega) \rightarrow 0$.

Heuristic for the decay. A surviving discriminant needs the ranks $\rho_p(D)$ of all $\approx \omega$ missing primes to have $\text{lcm} \leq L \approx p_\omega$. Modeling $\rho_p(D)$ as roughly uniform in $[1, p]$, the chance a single prime has small enough rank is of order L/p , and requiring many primes simultaneously multiplies these factors. The product falls faster than 2^ω grows, consistent with the observed net decay. This is heuristic: ranks are not independent, and a rigorous bound on the joint rank-LCM distribution is exactly what a proof requires.

5 The resulting open problem

The collapse suggests a route to unconditional termination that is combinatorial rather than analytic.

Conjecture 1 (Rank-LCM survival bound). *There is an explicit ω_0 such that for all $\omega \geq \omega_0$, no squarefree $D \mid P_\omega$ has both $y_1(D)$ p_ω -smooth and $R(D) = \text{lcm}\{\rho_p(D) : p \text{ missing}\} \leq L(\omega)$. Equivalently $S(\omega) = 0$ for $\omega \geq \omega_0$.*

Proposition 1. *Conjecture 1, together with the unconditional verification for $\omega < \omega_0$, implies that 633555×633556 is the last prime-complete product of two consecutive integers.*

Conjecture 1 is a statement about ranks of apparition in Pell–Lucas sequences—classical territory—rather than about the size of smooth pairs. It is the combinatorial counterpart of the analytic ceiling bound $\log A002072(\omega) = O(\omega)$ used elsewhere in this program; either would close the termination, and the rank-LCM form has the advantage of being directly computable (the solver prints $S(\omega)$) and monotone in the data.

Remark 2 (Status). The monotone decay is verified through $\omega = 17$; this is strong evidence, not a proof. A survivor cannot be excluded at some larger ω by a rank coincidence until Conjecture 1 is established. The natural next evidence is the continuation of $S(\omega)$ at $\omega = 18, 19, 20$ —where the fit projects S to fall below 1—and a rigorous bound on the proportion of $D \mid P_\omega$ with $R(D) \leq L$.

6 Implementation notes

The reference implementation (`LC_Solver`, v6.6) solves Pell equations through a `cypari2` fast path with a persistent subprocess `gp` fallback, distributes high-prime mask blocks across workers, and records per-order summaries (discriminant counts by gate, survivors, indices checked, hits) as JSON. Because D is generated squarefree, the Pell routine performs no internal factorization. The rank recurrence is iterated modulo each missing prime in machine integers; only surviving discriminants incur big-integer Pell-power work, and only at multiples of $R(D)$, where here and throughout $R(D)$ denotes the least common multiple of the relevant ranks of apparition. At $\omega = 16$ the entire order is decided in about 18 minutes on a single multi-core host, with three discriminants reaching an index check; $\omega = 17$, roughly five hours, leaves two. Checkpoint files persist the aggregated funnel statistics as well as the set of completed mask-blocks, so an interrupted order resumes with correct accounting and an accurate summary.